

EXPERIMENT :4

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1. Aim

- i) Consider a relation R having attributes as R(ABCD), functional dependencies are given below: $AB \rightarrow C$, $C \rightarrow D$, $D \rightarrow A$ Identify the set of candidate keys possible in relation R. List all the set of prime and non-prime attributes.
- ii) Relation R(ABCDE) having functional dependencies as : $A \rightarrow D$, $B \rightarrow A$, $BC \rightarrow D$, $AC \rightarrow BE$ Identify the set of candidate keys possible in relation R. List all the set of prime and non-prime attributes.
- iii) Consider a relation R having attributes as R(ABCDE), functional dependencies are given below: $B \rightarrow A$, $A \rightarrow C$, $BC \rightarrow D$, $AC \rightarrow BE$. Identify the set of candidate keys possible in relation R. List all the set of prime and non-prime attributes.
- iv) Consider a relation R having attributes as R(ABCDEF), functional dependencies are given below: $A \rightarrow BCD$, $BC \rightarrow DE$, $B \rightarrow D$, $D \rightarrow A$ Identify the set of candidate keys possible in relation R. List all the set of prime and non-prime attributes.
- v) Designing a student database involves certain dependencies which are listed below: $X \rightarrow Y$, $WZ \rightarrow X$, $WZ \rightarrow Y$, $Y \rightarrow W$, $Y \rightarrow X$, $Y \rightarrow Z$. The task here is to remove all the redundant FDs for efficient working of the student database management system.

2. Tools Used: SQL Server Management Studio

3. Query

Q1) Candidate keys: {AB, BC, BD}

Prime attributes: A, B, C, D

Non-prime attributes: none

Highest normal form:

- 1NF: satisfied (atomic attributes).
- 2NF: applies to partial dependencies of non-prime attributes on part of a candidate key. There are no non-prime attributes, so 2NF holds (vacuously).

- 3NF: For every FD $X \rightarrow Y$, either X is a superkey or Y is prime.
 - $AB \rightarrow C$: AB is a candidate key $\rightarrow$ OK.
 - $C \rightarrow D$: C is not a superkey, but D is prime $\rightarrow$ allowed in 3NF.
 - $D \rightarrow A$: D is not a superkey, but A is prime $\rightarrow$ allowed in 3NF. So 3NF holds.
- BCNF: Requires every FD's left side be a superkey. $C \rightarrow D$ and $D \rightarrow A$ have non-superkey left sides, so BCNF is violated.

Therefore, the highest normal form of R is 3NF (but not BCNF).

Q2) Candidate keys = $\{AC, BC\}$.

Prime attributes: A, B, C

Non-prime attributes: D, E

Highest normal form:

- 1NF: holds (attributes atomic).
- 2NF: requires no partial dependency of a non-prime attribute on a proper subset of a candidate key. Here $A \rightarrow D$: A is a proper subset of the candidate key AC , and D is non-prime. That is a partial dependency of a non-prime attribute $\rightarrow$ violates 2NF.
- Since 2NF is violated, higher normal forms (2NF, 3NF, BCNF) cannot hold.

Therefore, the highest normal form of R is 1NF.

Q3) Candidate keys: A and B

Prime attributes: A, B

Non-prime attributes: C, D, E

Highest normal form:

- 1NF: holds (attributes atomic).
- 2NF: holds. Partial-dependency concerns only arise when a proper part of a (composite) candidate key determines a non-prime attribute. Here candidate keys are single attributes (A and B), so there are no proper subsets to cause partial dependencies $\rightarrow$ 2NF holds.
- 3NF: For every FD $X \rightarrow Y$, either X is a superkey or each attribute of Y is prime. All FDs have left sides that are superkeys:
 - $B \rightarrow A$: B is a candidate key $\rightarrow$ OK.
 - $A \rightarrow C$: A is a candidate key $\rightarrow$ OK.
 - $BC \rightarrow D$: BC contains key B so is a superkey $\rightarrow$ OK.
 - $AC \rightarrow E$: AC contains key A so is a superkey $\rightarrow$ OK. So 3NF holds.
- BCNF: Requires every FD left side to be a superkey. As shown above, every FD's left side is a superkey.

Therefore, BCNF also holds. The relation is in BCNF (hence also in 3NF and 2NF).

Q4) Candidate keys: $\{AF, BF, DF\}$

Prime attributes: A, B, D, F

Non-prime attributes: C, E

Highest normal form:

- 1NF: satisfied (attributes atomic).
- 2NF: violated. Reason: 2NF forbids a partial dependency of a non-prime attribute on part of a composite key. Keys here are composite (size 2). For example, key AF — the part A (a proper subset) functionally determines C via $A \rightarrow C$ to $CA \rightarrow C$, and C is non-prime. That is a partial dependency of a non-prime attribute on part of a candidate key $\rightarrow$ violates 2NF.
- Since 2NF fails, higher normal forms (3NF, BCNF) do not hold.

Therefore, the highest normal form is: 1NF.

Q5) Candidate keys: {X, Y, WZ}

Prime attributes: W, X, Y, Z (all)

Non-prime: none

Highest normal form:

- 1NF: satisfied (attributes atomic).
- 2NF: there is one composite key WZ. Partial-dependency violations require a non-prime attribute depending on part of a composite key. All attributes are prime, so 2NF holds (vacuously).
- 3NF: for every FD $X \rightarrow A$, either X is a superkey or A is prime. Here every given FD has a left side that is a superkey (X and Y are keys; WZ is a key), so 3NF holds.
- BCNF: requires every FD left side to be a superkey. That is also true for the given FDs.

Therefore, the relation is in BCNF (and hence also in 3NF and 2NF).

Q6) Candidate key(s): AF P

Prime attributes: A, F

Non-prime attributes: B, C, D, E

Highest normal form:

- 1NF: holds (attributes are atomic).
- 2NF: requires that no non-prime attribute depend on a proper subset of a composite candidate key. Here the candidate key is composite {A,F}. We have $A \rightarrow BC$ and $A \rightarrow D$ (and thus $A \rightarrow E$), so the non-prime attributes B,C,D,E depend on A, which is a proper part of the key {A,F}. That is a partial dependency $\rightarrow$ violates 2NF.
- Since 2NF is violated, the relation cannot be in 3NF or BCNF.

Therefore, the highest normal form is: 1NF.



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4. Learning Outcomes

- a) Learned to compute candidate keys using attribute closure.
- b) Understood how to classify prime and non-prime attributes.
- c) Identified partial dependencies and their effect on normalization.
- d) Gained practical insight into reducing redundancy and anomalies in database design.